Advanced Machine Learning

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Recap – intractability of learning

Main Result: *There exist classes that are PAC learnable for which the sample complexity is polynomial but the runtime is not polynomial.*

Consider $\mathcal{H}_{3DNF}^{d}$ = class of 3-term disjunctive normal form formulae consisting of hypothesis of the form h: $\{0,1\}^{d} \rightarrow \{0,1\}$,

$$h(\mathbf{x}) = A_1(\mathbf{x}) \vee A_2(\mathbf{x}) \vee A_3(\mathbf{x}),$$

where each term $A_i(\mathbf{x})$ is a Boolean conjunction (in $\mathcal{H}_{conj}^{d}$) with at most d Boolean literals $x_1, \ldots, x_d$.

$|\mathcal{H}_{3DNF}^{d}| = 3^{3d} < \infty$, so is PAC learnable with sample complexity 3d $\log(3/\delta)/\varepsilon$ (polynomial in $1/\varepsilon$, $1/\delta$, d).

Let $\mathcal{H}$ be a hypothesis class that is efficient PAC learnable. Then, there exists a randomized algorithm which solves the problem of finding a hypothesis in $\mathcal{H}$ consistent with a given training sample, and which has runtime polynomial in m (the length of the training sample).

Recap – intractability of learning

The class $\mathcal{H} = \mathcal{H}_{3DNF}^{n}$ is PAC learnable but from the computational perspective, the learning problem is hard. There is no polynomial time algorithm (unless RP = NP) that *properly* learns from training data. So ERM_H is not efficient. If $\mathcal{H}_{3DNF}^{d}$ is efficient PAC learnable then the problem of graph 3-coloring problem (which is shown to be NP-complete) is in RP.

So, the class $\mathcal{H}_{3DNF}^{n}$ is not efficient PAC learnable under the assumption that NP-complete problems cannot be solved with high probability by a probabilistic polynomial–time algorithm (technically, under the assumption RP $\neq$ NP).

- "Consistent learning is hard" $\not\Rightarrow$ "learning is hard"

- "Consistent learning is hard" $\Rightarrow$ '*proper* learning is hard"
- A proper learning algorithm is a learning algorithm that must output $h \in \mathcal{H}$

Improper learning of 3-term DNFs

Use the distribution rule to obtain:
$(a \wedge b) \vee (c \wedge d) = (a \vee c) \wedge (a \vee d) \wedge (b \vee c) \wedge (b \vee d)$

Apply for the 3-term DNF formula to obtain a 3-CNF formulae:

each u,v,w can take 2n values in

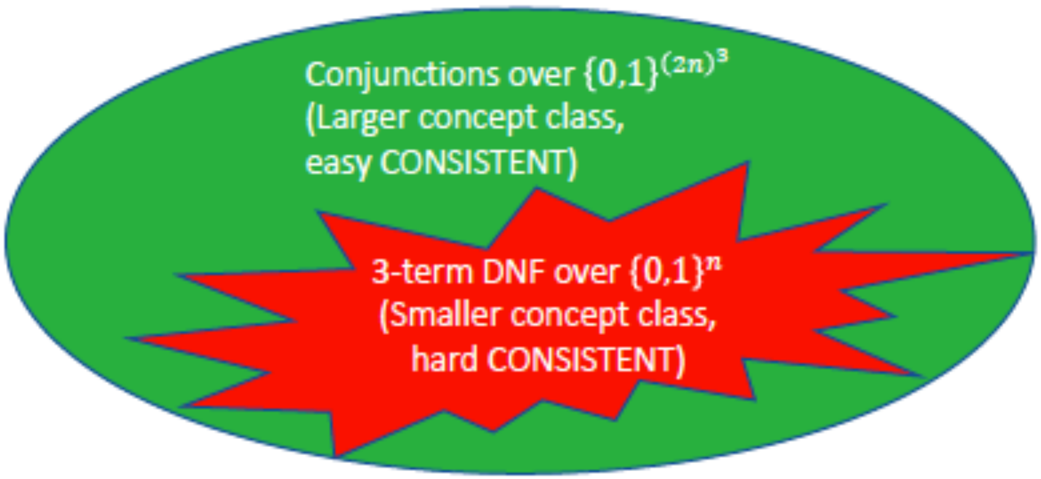
$$A_1 \vee A_2 \vee A_3 = \bigwedge_{u \in A_1, v \in A_2, w \in A_3} (u \vee v \vee w) = \bigwedge_{u \in A_1, v \in A_2, w \in A_3} y_{u,v,w} \qquad \{x_1, \overline{x_1}, x_2, \overline{x_2}, \ldots, x_n, \overline{x_n}\}$$

So we have that a 3-term DNF can be viewed as a conjunction of $(2n)^3$ variables:

$$H_{3DNF}^{n} \subseteq H_{conj}^{(2n)^3} \qquad \left| H_{conj}^{(2n)^3} \right| = 3^{(2n)^3} + 1$$

We can efficiently PAC learn the new class of conjunctions, $H_{conj}^{(2n)^3}$ with sample complexity $(n^3+\log(1/\delta))/\varepsilon$. The overall runtime of this approach is polynomial in $1/\varepsilon$, $1/\delta$, n. Pay polynomially in sample complexity, gain exponentially in computational complexity.

Improper learning



	3-term DNF	Conjunctions over $\{0,1\}^{(2n)^3}$	
Sample complexity	$O\left(\frac{n+\log\frac{1}{\delta}}{\varepsilon}\right)$	$O\left(\frac{n^3+\log\frac{1}{\delta}}{\varepsilon}\right)$	⬆
CONSISTENT Computational-Complexity	NP-Hard	$O\left(n^3 \times \frac{n^3+\log\frac{1}{\delta}}{\varepsilon}\right)$	⬇

Hardness of learning

Some classes are hard to PAC learn if we place certain restrictions on the hypothesis class used by the learning algorithm

- the problem of properly learning 3-term DNF formulae is computationally hard (the learning algorithm is restricted to output a hypothesis of the class $\mathcal{H}_{3\mathrm{DNF}}{}^{\mathrm{d}}$)
- this problems can be solved efficiently by leting the learning algorithm to output a hypothesis from the class $H_{conj}^{(2n)^3}$
- representation dependent hardness

Interesting and fundamental question regarding the PAC learning model:

- are there any classes that are computationally hard to learn, independent of the representation used?
- we are interested in the existence of classes with polynomial VC dimension (such that we need polynomial number of training examples to PAC learn them – thus is no information-theoretic barrier to fast learning), yet there is no algorithm with runtime polynomial.
- how to prove that a class is computational hard, independent of its representation?

Learning vs. cryptography

In some sense, cryptography is the opposite of learning:

- in learning we try to uncover some rule underlying the examples we see
- in cryptography, the goal is to make sure that nobody will be able to discover some secret, in spite of having access to some partial information about it.

Results about the cryptographic security of some system translate into results about the un-learnability of some corresponding task. The common approach for proving that cryptographic protocols are secure is to start with some cryptographic assumptions. It is our belief that they really hold.

Deduce hardness of learnability from cryptographic assumptions.

Use the concepts of one way function (*easy to compute, hard to invert*) + trapdoor one way function *(although is hard to invert, once one has access to its secret key, inverting becomes feasible) to build the class* $\mathcal{H}_F^n = \{ f^{-1} : f$ is a trapdoor function$\}$ with polynomial VC dimension but hard to learn.

Learning vs. cryptography

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Usual setup: $S = \{(x_i, f(x_i))\}$.
For finding inverse function f^{-1} : take $S = \{(f(x_i), x_i)\}$. Is hard to lean the invers function f^{-1} such that $f^{-1}((f(x_i)) = x_i$

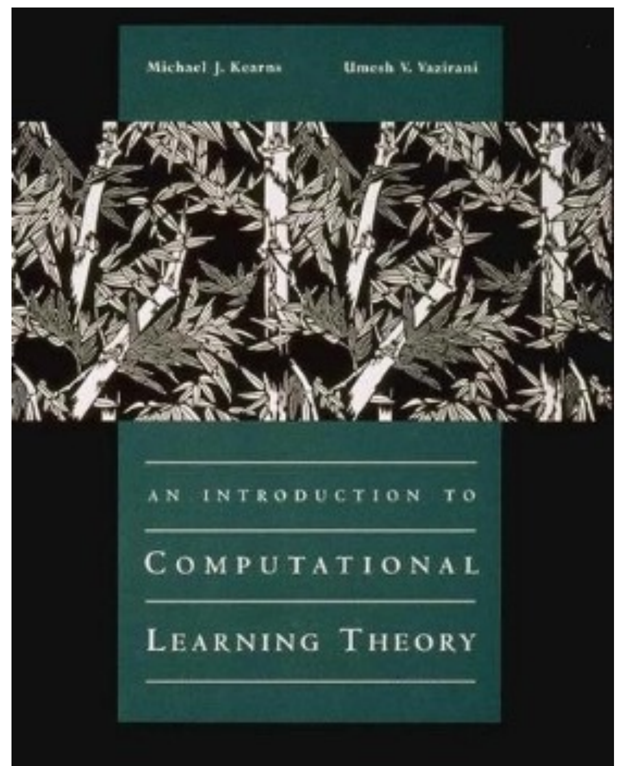
Other examples

Discrete Cube Root Problem. Let p and q be two primes. And 3 does not divide $(p-1)(q-1)$. Let $N = pq$. Then Let $x \in Z^*$ and $x \leq N$. Let $f_N(x) = x^3 \ mod \ N$. The problem is that given N and $y = f_N(x)$, find x.

For any polynomial $p(n)$, there is no algorithm such that,

1. That runs in time $p(n)$ and
2. On input $N = pq$ from two randomly chosen n-bit primes p and q, such that 3 does not divide $(p-1)(q-1)$ and input $y \neq 0$ s.t. neither p nor q can divide y, chose uniformly at random and returns x, such that $x^3 \ mod \ N = y$. w.p$\geq \frac{1}{p(n)}$ (over p, q, y and algorithm choices)

Kearns & Vazirani, MIT Press:
An Introduction to Computational Learning Theory



Boosting

Overview of Boosting

Boosting = general method of converting simple (weak) classifiers (better than chance) into highly accurate prediction rule

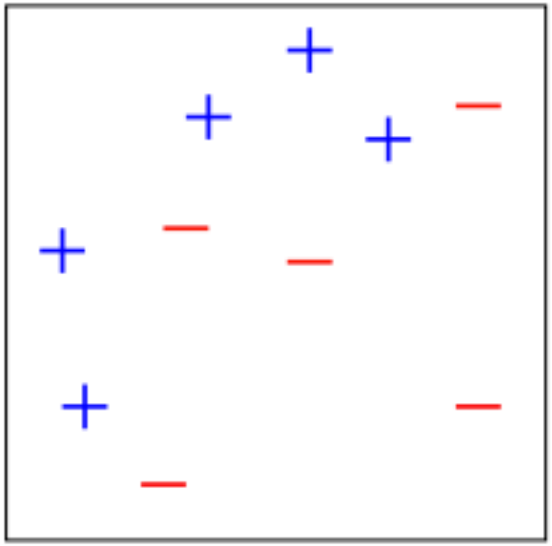
Main idea of Boosting:

- assume given "weak" learning algorithm that uses a simple "rule of thumb" to output a hypothesis that comes from an easy-to-learn hypothesis class and performs just slightly better than a random guess ($0 < \varepsilon < 0.5$)

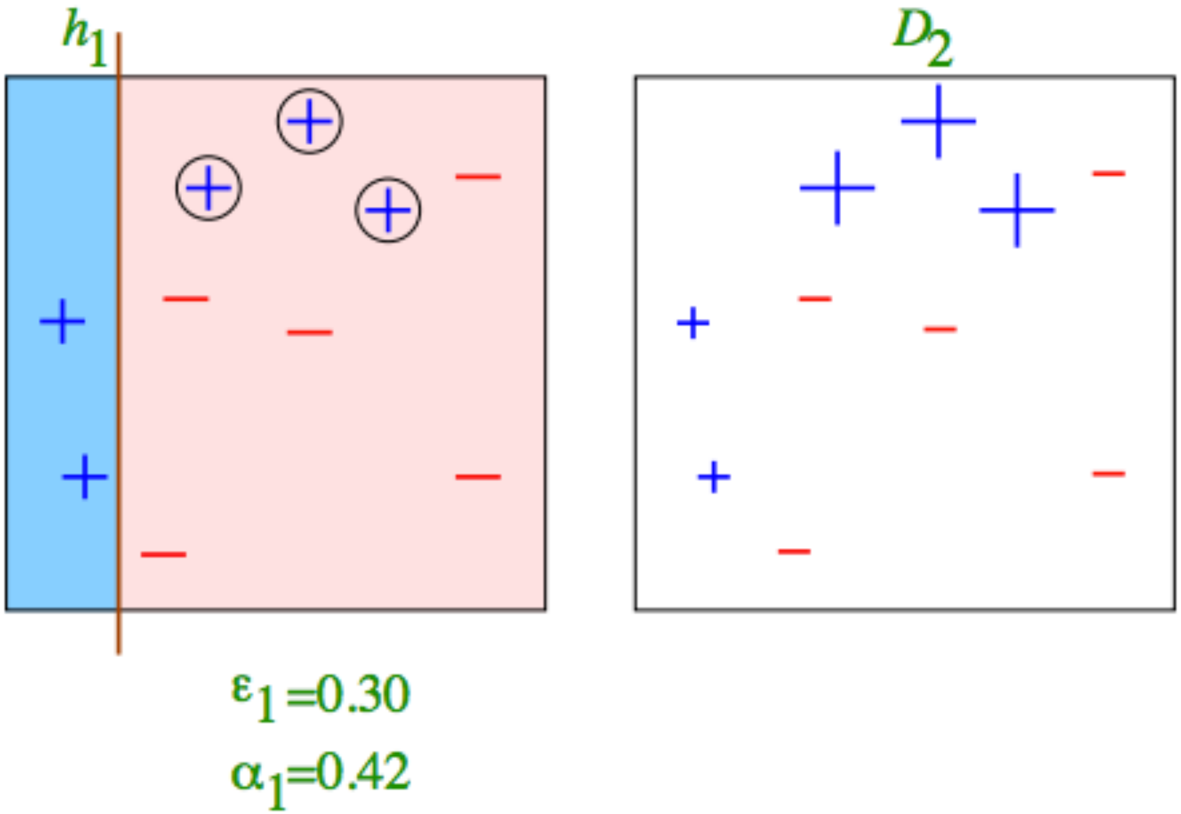
- a boosting algorithm amplifies the accuracy of weak learners by aggregating such weak classifiers to approximates gradually good predictors for larger, and complex, classes.

Boosting is a great example for the practical impact of learning theory. While boosting originated as a purely theoretical problem, it has led to popular and widely used algorithms. It has been successfully used for learning to detect faces in images.

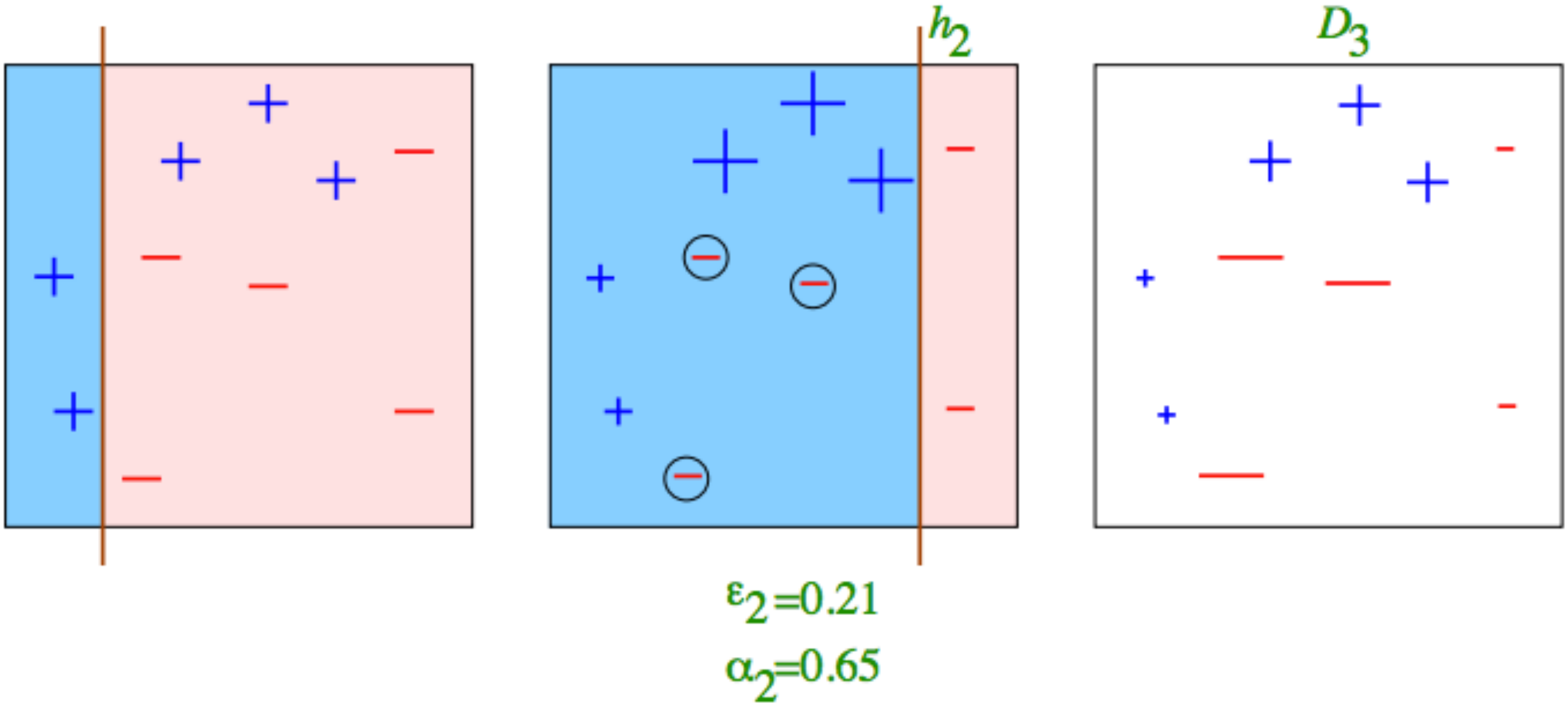
Toy example



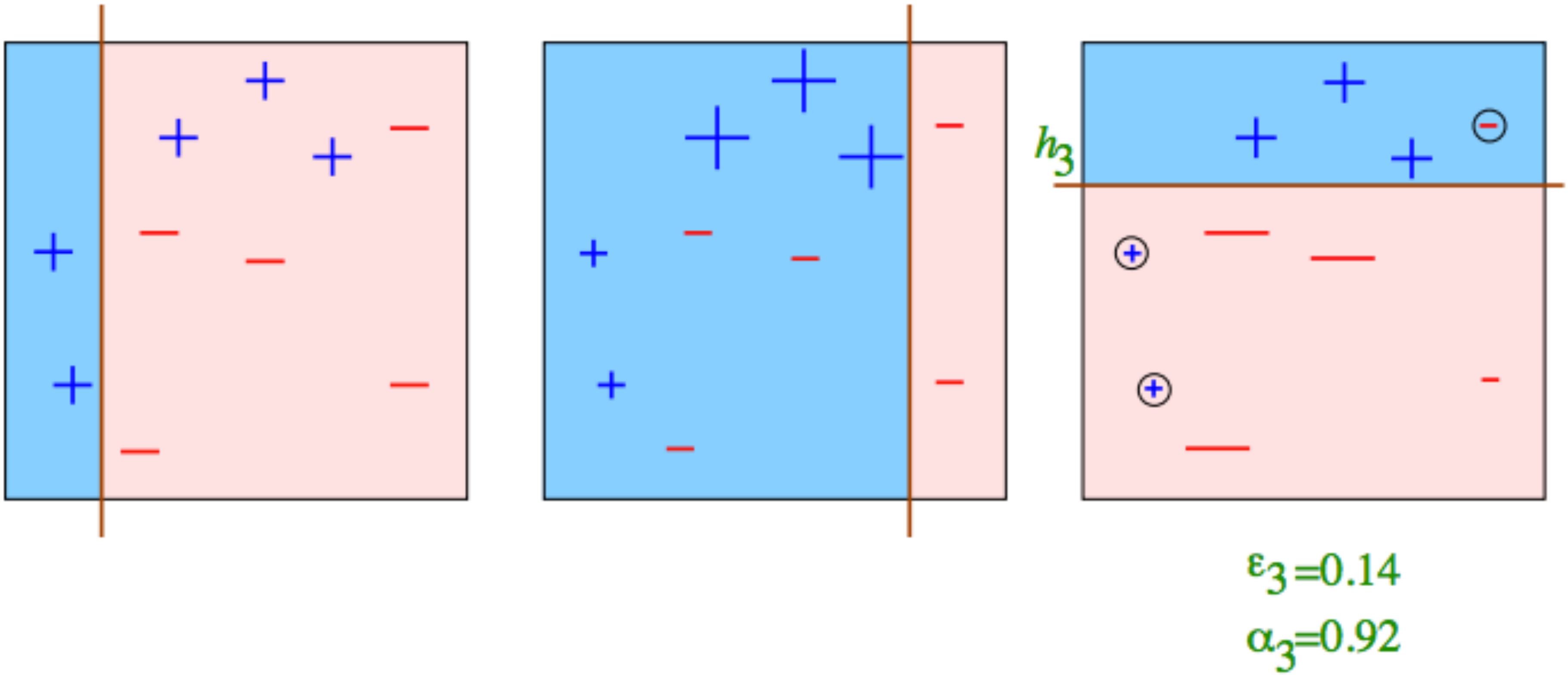
Toy example



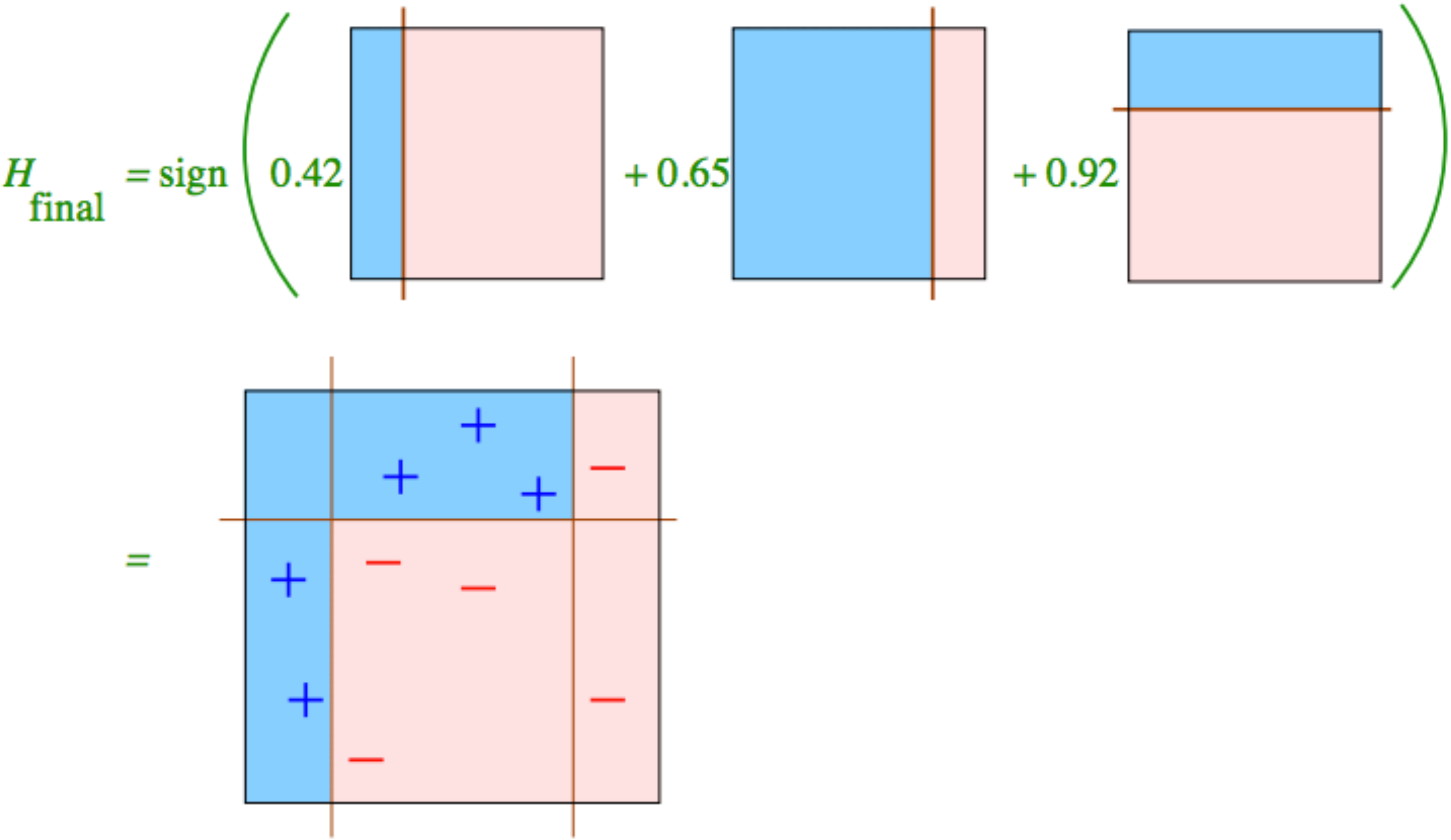
Toy example



Toy example



Toy example – final classifier



Next week: Viola-Jones face detector

ACCEPTED CONFERENCE ON COMPUTER VISION AND PATTERN RECOGNITION 2001

Rapid Object Detection using a Boosted Cascade of Simple Features

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Abstract

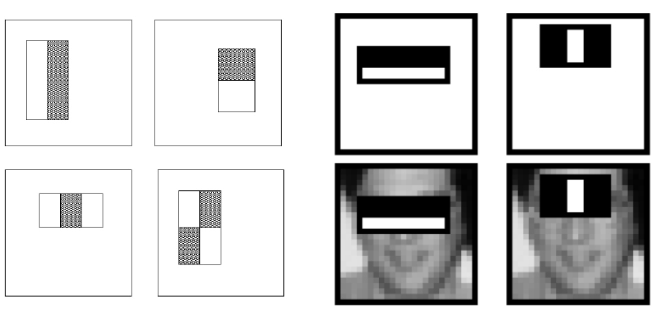
This paper describes a machine learning approach for visual object detection which is capable of processing images

tected at 15 frames per second on a conventional 700 MHz Intel Pentium III. In other face detection systems, auxiliary information, such as image differences in video sequences, or pixel color in color images, have been used to achieve

Viola-Jones face detector

Main idea:

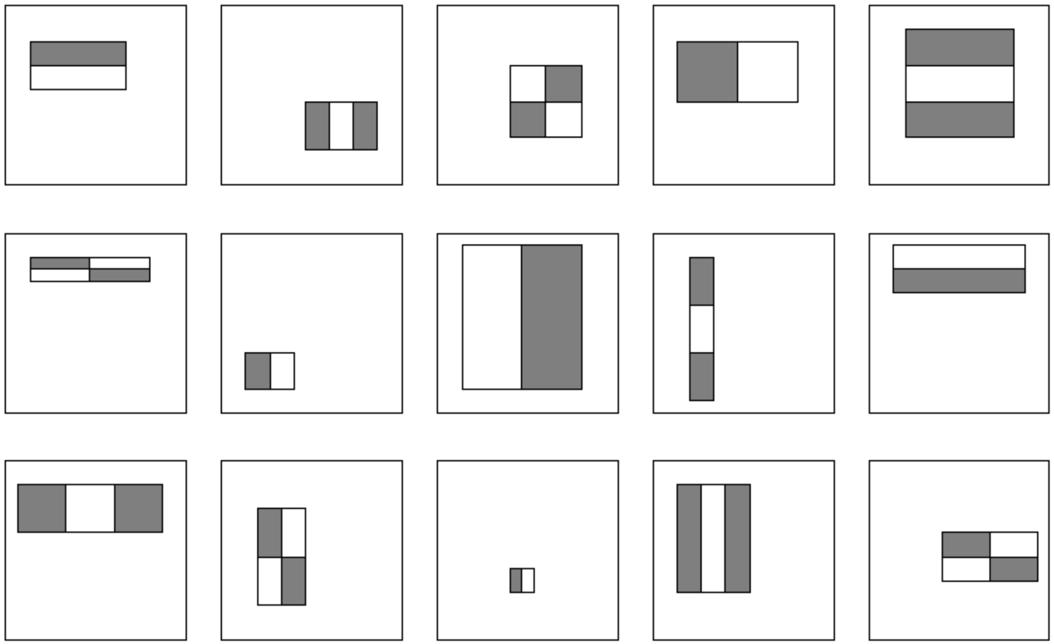
- represent local texture with efficiently computable "rectangular" features within window of interest
- select discriminative features to be weak classifiers
- use boosted combination of them as final classifier
- form a cascade of such classifiers, rejecting clear negatives quickly



"Rectangular" filters

Feature output is difference between adjacent regions

Viola-Jones face detector: features



Considering all possible filter parameters: position, scale, and type:

180,000+ possible features associated with each 24 × 24 window

Which subset of these features should we use to determine if a window has a face?

Use AdaBoost both to select the informative features and to form the classifier

Viola-Jones face detector: features

Figure 3: The first and second features selected by AdaBoost. The two features are shown in the top row and then overlayed on a typical training face in the bottom row. The first feature measures the difference in intensity between the region of the eyes and a region across the upper cheeks. The feature capitalizes on the observation that the eye region is often darker than the cheeks. The second feature compares the intensities in the eye regions to the intensity across the bridge of the nose.

For example an excellent first stage classifier can be constructed from a two-feature strong classifier by reducing the threshold to minimize false negatives. Measured against a validation training set, the threshold can be adjusted to detect 100% of the faces with a false positive rate of 40%. See Figure 3 for a description of the two features used in this classifier.

Computation of the two feature classifier amounts to about 60 microprocessor instructions. It seems hard to imagine that any simpler filter could achieve higher rejection rates. By comparison, scanning a simple image template, or a single layer perceptron, would require at least 20 times as many operations per sub-window.

PAC Learnability = Strong learnability

Remember the definition of a class $\mathcal{H}$ being PAC Learnable:

A hypothesis class $\mathcal{H}$ is called ***PAC learnable*** if there exists a function $m_{\mathcal{H}}$: $(0,1)^2 \to N$ and a learning algorithm A with the following property:

- for every $\varepsilon > 0$ (*accuracy* → *"approximately correct"*)
- for every $\delta > 0$ (*confidence* → *"probably"*)
- for every labeling $f \in \mathcal{H}$ (*realizability case*)
- for every distribution $\mathcal{D}$ over $\mathcal{X}$

when we run the learning algorithm A on a training set, consisting of $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$ examples sampled i.i.d. from $\mathcal{D}$ and labeled by f the algorithm A returns a hypothesis $h \in \mathcal{H}$ such that, with probability at least $1-\delta$ (over the choice of examples), $L_{\mathrm{D},f}(h) \leq \varepsilon$.

Given sufficiently many examples we can learn a classifier from $\mathcal{H}$ with arbitrary small generalization error ε and with arbitrary high confidence $1-\delta$.
Strong learnability = ability to learn a classifier with arbitrary small generalization error

Weak learnability

Definition (γ-Weak-Learnability)
A learning algorithm, A, is a γ-weak-learner for a class $\mathcal{H}$ if there exists a function $m_{\mathcal{H}}:(0,1)\rightarrow N$ such that:

- for every $\delta > 0$ (*confidence*)
- for every labeling $f \in \mathcal{H}, f : \mathcal{X} \rightarrow \{-1,+1\}$ (*realizability case*)
- for every distribution $\mathcal{D}$ over $\mathcal{X}$

when we run the learning algorithm A on a training set, consisting of $m \geq m_{\mathcal{H}}(\delta)$ examples sampled i.i.d. from $\mathcal{D}$ and labeled by f, the algorithm A returns a hypothesis h (h might not be from $\mathcal{H}$ - improper learning) such that, with probability at least $1-\delta$ (over the choice of examples), $L_{\mathcal{D},f}(h) \leq 1/2 - \gamma$.

A hypothesis class $\mathcal{H}$ is γ-weak-learnable if there exists a γ-weak-learner for that class.

Weak vs Strong learnability

Strong learnability implies the ability to find an arbitrarily good classifier (with error rate at most ε for an arbitrarily small $\varepsilon > 0$).

In weak learnability, however, we only need to output a hypothesis whose error rate is at most $1/2-\gamma$, namely, whose error rate is slightly better than what a random labeling would give us.

The hope is that it may be easier to come up with ***efficient*** weak learners than with efficient (strong) PAC learners. If we have access to an ***efficient*** weak learner, can we use it to build an ***efficient*** strong learner?

We will see that strong learnability $\Leftrightarrow$ weak learnability

$\Rightarrow$ easy to show, based on the definition

$\Leftarrow$ use boosting (improper learning algorithm) that combines weak learners to obtain a strong learner.

If the learning problem is hard, boosting cannot help as we can't find efficient weak learners.

Weak vs Strong learnability

The fundamental theorem of learning (lecture 8) states that if a hypothesis class $\mathcal{H}$ has a VC dimension d, then the sample complexity of PAC learning $\mathcal{H}$ satisfies

$$C_1 \frac{d + \log(1/\delta)}{\epsilon} \leq m_{\mathcal{H}}(\epsilon, \delta) \leq C_2 \frac{d \log(1/\epsilon) + \log(1/\delta)}{\epsilon}$$

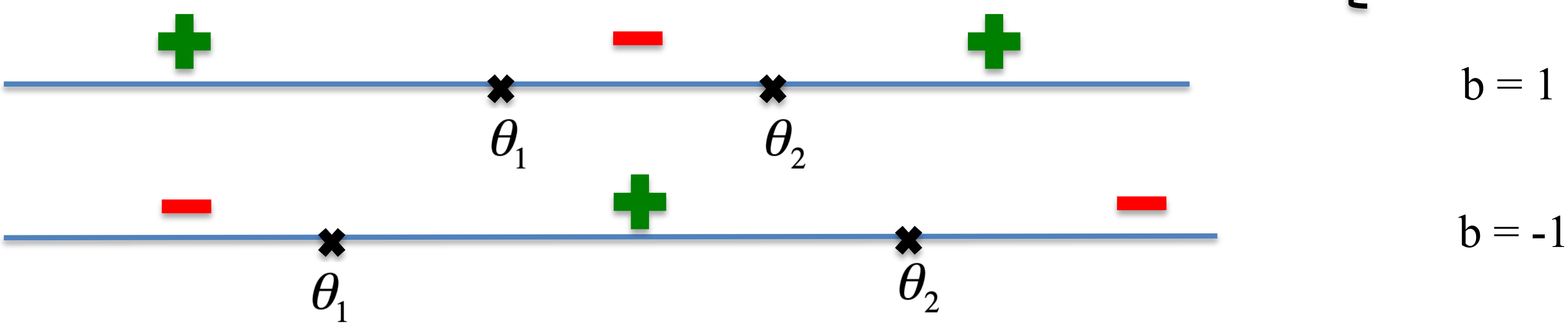
Applying this with $\varepsilon = 1/2 - \gamma$ we immediately obtain that if $d = \infty$ then $\mathcal{H}$ is not γ -weak-learnable. This implies that ***from the statistical perspective*** (i.e., if we ignore computational complexity), ***weak learnability*** is also characterized by the VC dimension of $\mathcal{H}$ and therefore ***is just as hard as PAC (strong) learning.***

However, when we do consider computational complexity, the potential advantage of weak learning is that maybe there is an algorithm that satisfies the requirements of weak learning and can be implemented efficiently.

Weak learnability - example

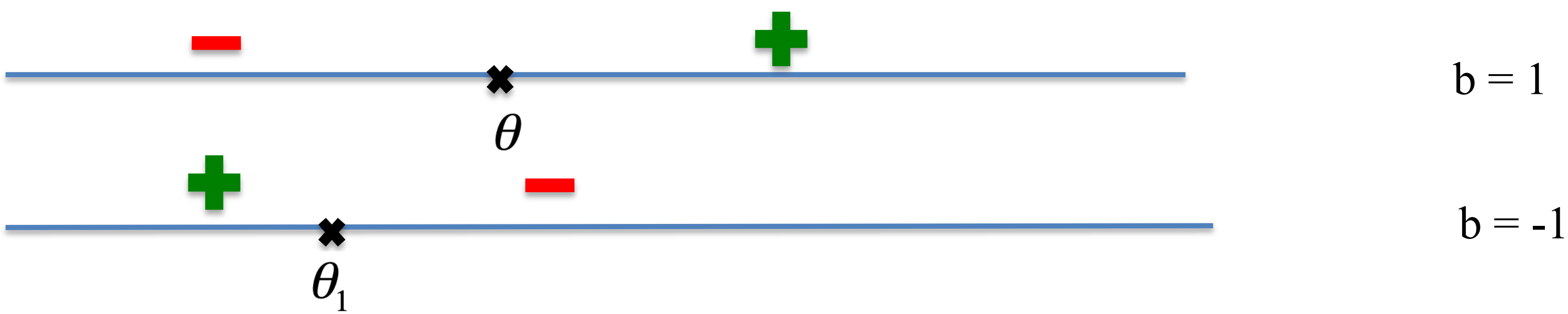
Let $\mathcal{X} = \mathrm{R}$, $\mathcal{H}$ is the class of 3-piece classifiers (signed intervals):

$$H = \{h_{\theta_1,\theta_2,b} \mid \theta_1, \theta_2 \in R, \theta_1 < \theta_2, b \in \{-1,+1\}\} \quad h_{\theta_1,\theta_2,b}(x) = \begin{cases} +b, \text{ if } x < \theta_1 \text{ or } x > \theta_2 \\ -b, \text{ if } \theta_1 \le x \le \theta_2 \end{cases}$$



Consider $\mathcal{B}$ the class of Decision Stumps = class of 1-node decision trees

$\mathcal{B} = \{h_{\theta,b}: \mathbf{R} \rightarrow \{-1,1\}, h_{\theta,b}(x) = \mathrm{sign}(x - \theta) \times b, \theta \in \mathbf{R}, b \in \{-1,+1\}\}$.

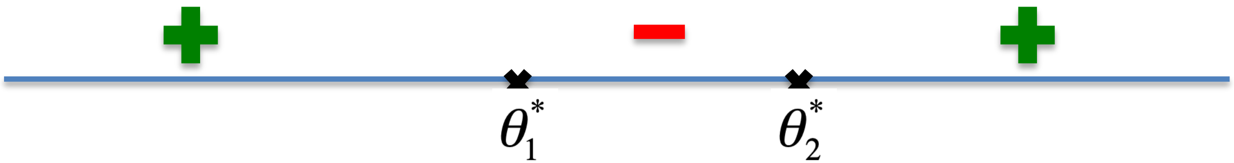


$\mathrm{ERM}_{\mathcal{B}}$ is a γ-weak learner for $\mathcal{H}$, for $\gamma = 1/12$.

Weak learnability - example

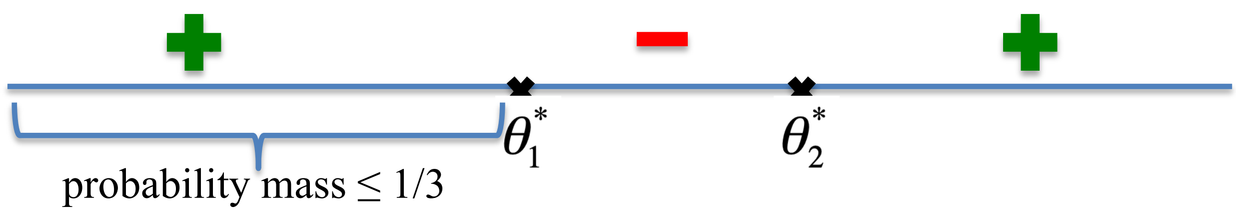
$ERM_{\mathcal{B}}$ is a γ-weak learner for $\mathcal{H}$, for $\gamma = 1/12$.

Proof: Consider a $h^* = h_{\theta_1^*, \theta_2^*, b^*} \in \mathcal{H}$ that labels a training set S.

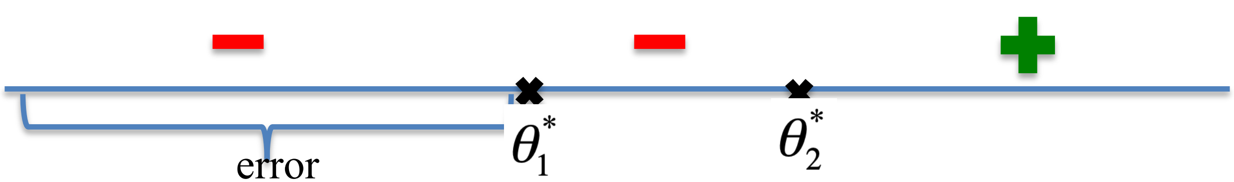


Consider a distribution $\mathcal{D}$ over $\mathcal{X} = \mathbf{R}$. Then, we are sure that at least one of the regions $(-\infty, \theta_1^*)$, $[\theta_1^*, \theta_2^*]$, $(\theta_2^*, +\infty)$ has a probability mass wrt $\mathcal{D} \leq 1/3$.

Consider, without loss of generality that $\mathcal{D}((-\infty, \theta_1^*)) = P_{x \sim \mathcal{D}}(x \in (-\infty, \theta_1^*)) \leq 1/3$
Then the hypothesis $h_{\theta,b} \in B$, where $\theta = \theta_2^*$, $b = b^*$ errors on $(-\infty, \theta_1^*)$.



$h_{\theta_1^*, \theta_2^*, b^*} \in \mathcal{H}$



$\in \mathcal{B}$

Weak learnability - example

$\mathcal{B} = \{h_{\theta,b}:\mathbf{R}\rightarrow\{-1,1\},\ h_{\theta,b}(x) = \text{sign}(x - \theta) \times b,\ \theta \in \mathbf{R},\ b \in \{-1,+1\}\}$.

It is easy to show that VCdim($\mathcal{B}$) = VCdim(class of signed thresholds) = 2.

The fundamental theorem of learning states that if the hypothesis class $\mathcal{B}$ has VCdim($\mathcal{B}$) = d, then the sample complexity of agnostic PAC learning $\mathcal{B}$ satisfies:

$$C_1 \frac{d + \log(1/\delta)}{\epsilon^2} \leq m_{\mathcal{H}}(\epsilon, \delta) \leq C_2 \frac{d + \log(1/\delta)}{\epsilon^2}$$

So, if the sample size is greater than the sample complexity, then with probability of at least 1 – δ, the $\text{ERM}_{\mathcal{B}}$ rule learns in the agnostic case a hypothesis such that:

$$L_{\mathcal{D}}(\text{ERM}_{\text{B}}(S)) \leq \min_{h\in\mathcal{H}} L_{\mathcal{D}}(h) + \epsilon \ = 1/3 + \varepsilon$$

Take $\varepsilon = 1/12$ than we obtain $1/3 + 1/12 = 5/12 \leq 1 - 1/12$. Than $\text{ERM}_{\mathcal{B}}$ is a γ-weak learner for $\mathcal{H}$, where $\gamma = 1/12$.

Efficient implementation of ERM for Decision Stumps

In practice, we use the following base hypothesis class of decision stumps over $\mathbf{R}^d$ for weak learners:

$\mathcal{H}_{DS}{}^d = \{h_{i,\theta,b}: \mathbf{R}^d \rightarrow \{-1,1\},\ h_{i,\theta,b}(\mathbf{x}) = \mathrm{sign}(\theta - x_i) \times b,\ 1 \leq i \leq d,\ \theta \in \mathbf{R},\ b \in \{-1,+1\}\}$

-pick a coordinate i (from 1 to d), project the input $x = (x_1, x_2, \ldots, x_d)$ on the i-th coordinate and obtain x_i, if $x_i \leq$ threshold θ label the example with b, else with –b

How to implement efficient ERM rule for the class $\mathcal{H}_{DS}{}^d$?

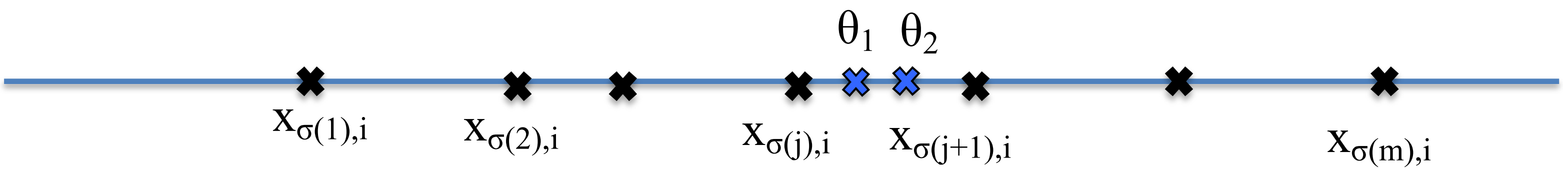
Let $S = ((\mathbf{x_1}, y_1), \ldots, (\mathbf{x_m}, y_m))$ be a training set of size m. We want to find the best h_{i^*,θ^*,b^*} which minimizes the training error on S:

$$h_{i^*,\theta^*,b^*} = \operatorname*{argmin}_{h_{i,\theta,b} \in H_{DS}^d} L_S(h_{i,\theta,b}) = \operatorname*{argmin}_{\substack{1 \leq i \leq d \\ \theta \in R \\ b \in \{-1,+1\}}} L_S(h_{i,\theta,b})$$

Efficient implementation of ERM for Decision Stumps

We have $1 \leq i \leq d$, $\theta \in \mathbf{R}$, $b \in \{-1,+1\}$. We fix $i \in \{1, 2, \ldots, d\}$ and $b \in \{-1,+1\}$. Then we are interested in minimizing the error on $S_i = ((x_{1,i}, y_1), \ldots, (x_{m,i}, y_m))$. By sorting $x_{1,i}, x_{2,i}, \ldots, x_{m,i}$ we obtain $x_{\sigma(1),i} \leq x_{\sigma(2),i} \leq \ldots \leq x_{\sigma(m),i}$

We have that $\theta \in \mathbf{R}$. Pick θ_1 and θ_2 in $[x_{\sigma(j),i}, x_{\sigma(j+1),i})$



We see that $h_{i,\theta 1,b}$ and $h_{i,\theta 2,b}$ have the same error. For all $\theta \in [x_{\sigma(j),i}\ x_{\sigma(j+1),i})$ we obtain the same error, all the hypothesis $h_{i,\theta,b}$ are very similar.

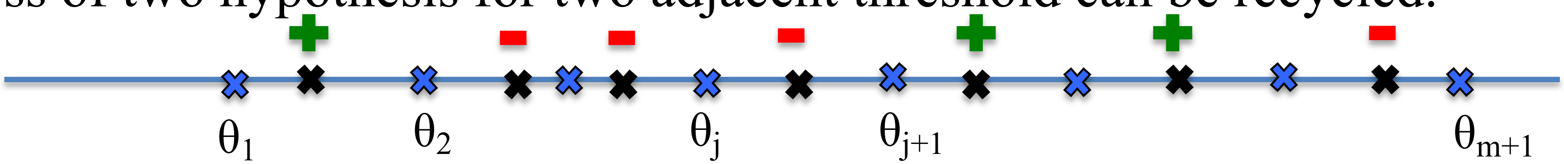
In fact, we can restrict the search over all $\theta \in \mathbf{R}$ just to $m + 1$ values of θ, chosen in the intervals $(-\infty, x_{\sigma(1),i})$, $[x_{\sigma(1),i}, x_{\sigma(2),i})$, $\ldots$, $[x_{\sigma(m),i}, +\infty)$ by taking a representative threshold in each interval. For example we can take the set of representative thresholds as containing extreme points + middle of the segments:
$\Theta_i = \{x_{\sigma(1),i}-1,\ 1/2 \times (x_{\sigma(1),i} + x_{\sigma(2),i}), \ldots, 1/2 \times (x_{\sigma(m-1),i} + x_{\sigma(m),i}),\ x_{\sigma(m),i}+1\}$

Efficient implementation of ERM for Decision Stumps

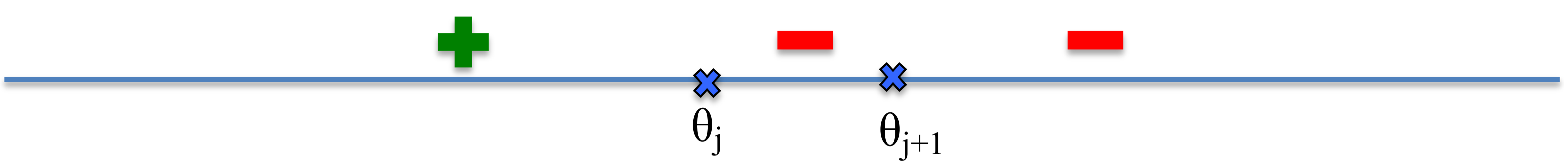
$\mathcal{H}_{DS}{}^{d} = \{h_{i,\theta,b}: \mathbf{R}^d \rightarrow \{-1,1\},\ h_{i,\theta,b}(\mathbf{x}) = \mathrm{sign}(\theta - x_i) \times b,\ 1 \le i \le d,\ \theta \in \mathbf{R},\ b \in \{-1,+1\}\}$

We have $1 \le i \le d$, $\theta \in \Theta_i$, $|\Theta_i| = m+1$, $b \in \{-1,+1\}$. So we have $d \times (m+1) \times 2$ possible hypothesis. Each hypothesis takes $O(m)$ runtime. So the runtime is polynomial.

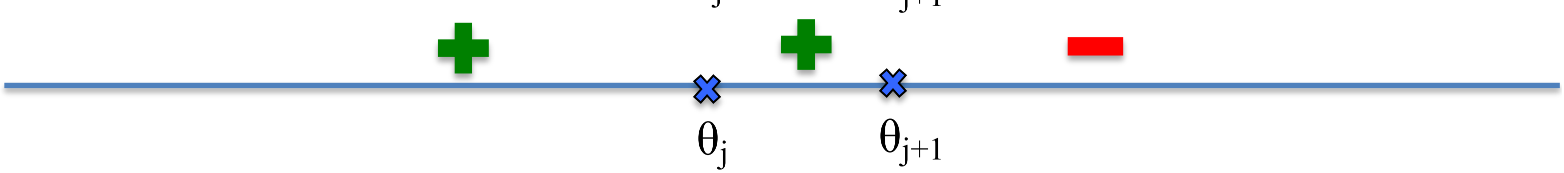
You can decrease the entire runtime using dynamic programming as computing the loss of two hypothesis for two adjacent threshold can be recycled.



Suppose b = 1. Then $L_S(h_{i,\theta_{j+1},1}) = L_S(h_{i,\theta_j,1}) - \frac{1}{m} y_j$



$h_{i,\theta_j,1} \in \mathcal{H}_{DS}{}^{d}$



$h_{i,\theta_{j+1},1} \in \mathcal{H}_{DS}{}^{d}$

Efficient implementation of ERM for Decision Stumps

$\mathcal{H}_{DS}{}^{d} = \{h_{i,\theta,b}: \mathbf{R}^d \rightarrow \{-1,1\},\ h_{i,\theta,b}(\mathbf{x}) = \text{sign}(\theta - x_i) \times b,\ 1 \le i \le d,\ \theta \in \mathbf{R},\ b \in \{-1,+1\}\}$

input: training set $S = ((\mathbf{x_1}, y_1), \ldots, (\mathbf{x_m}, y_m))$ of size *m*
goal: find $h_{i*,\theta*,b*}$ which minimizes the training error on S
initialize: $L^* = +\infty$
for $b = -1,+1$
    for $i = 1, \ldots, d$
        sort S on the i-th coordinate, obtain $x_{\sigma(1),i} \le x_{\sigma(2),i} \le \ldots \le x_{\sigma(m),i}$
        take $\Theta_i = \{\theta_1, \theta_2, \ldots, \theta_{m+1}\} = \{x_{\sigma(1),i}-1, \ldots 1/2 \times (x_{\sigma(j),i} + x_{\sigma(j+1),i}), \ldots, x_{\sigma(m),i}+1\}$
        compute current loss $L_{current} = L_S(h_{i,\theta_1,b})$
        if $L_{current} < L^*$
            $L^* = L_{current}$, $i^* = i$, $\theta^* = \theta_1$, $b^* = b$
        for $j = 1, \ldots, m$
            compute $L_{current} = L_S(h_{i,\theta_{j+1},b}) = L_S(h_{i,\theta_j,b}) - \frac{1}{m} b y_j$
            if $L_{current} < L^*$
                $L^* = L_{current}$, $i^* = i$, $\theta^* = \theta_1$, $b^* = b$
output $h_{i*,\theta*,b*}$
Runtime: $O(2 \times d \times (m \times \log_2 m + m)) = O(d \times m \times \log_2 m)$